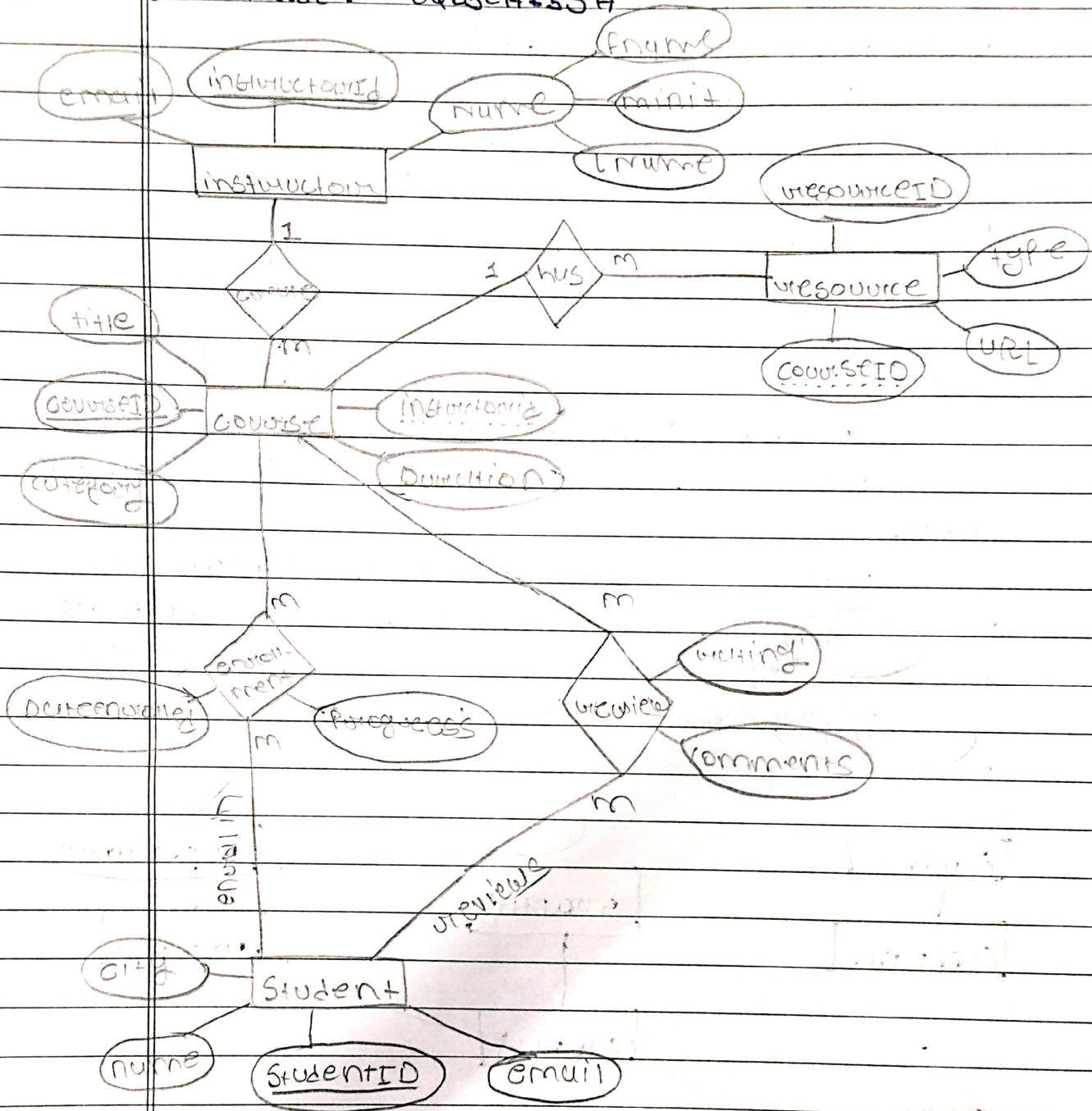


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SRN :- PES1PG25CA044

Subject code :- UQ25CA453A



1. map strong entities.

-> Primary key:

instructorid

courseid

courseid

studentid

-> Foreign key

instructorid -> create relationship (1:M)

courseid -> has relationship (1:M)

studentid, courseid -> enrollment relationship (M:M)
and review relationship (M:M)

-> composite attributes:

-> name -> fname, mname, lname

2 map weak entities

enrollment -> studentid, courseid

review -> studentid, courseid

3 map 1:1 relationship

in this scenario don't have any one to one relationship

4 map 1:M relationship

• each course is created by instructor
and createdid to course ~~has~~ coursePK = courseid FK = instructorid reference to
instructor (instructor)

Teacher's Signature:

- each resource has one course

ADD ~~resourceid~~ ^{courseid} in ~~course resource~~

PK = ~~courseid~~ ^{resourceid} FK = ~~resourceid~~ ^{reference}

~~resource (resourceid)~~
course (courseid)

5. map m:n relationship

→ enrollment (studid, courseid, dateenrolled, progress)

PK (studid, courseid)

FK₁ = studid → student (studid)

FK₂ = courseid → course (courseid)

→ review (rating, comments, studentid, courseid)

PK = studentid, courseid

FK₁ = student (studentid)

FK₂ = course (courseid)

6. map multivalued attributes.

in this scenario don't have any multivalued attributes.

7. map specialisation / generalisation

in this scenario don't have any specialisation / generalisation

→ the final list

→ instructor / instructorID

email

name

→ course : courseID

title

category

duration

resources (FK)

instructorid (FK instructor)

→ student : studentID

name

email

city

→ resource : resourceID

type

URL

type

courseid (FK)

→ enrollment : studentID (FK)

courseID (FK)

dateenrolled

progress

→ review: studentID (FK)

courseID (FK)

rating

comment

Teacher's Signature: